ASEN 6062: Celestial Mechanics
Summary of Important Results/Relations for the Point Mass N-Body Problem
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In the following we provide a series/summary of the most important results for the N-Body Problem and the Full Body Problem. These relations are not meant to be definitive, but are pulled from the lecture notes as important conditions or results.

The N-Body Problem

Given N particles $P_i, i = 1, 2, \ldots, N$, each with a defined mass, position and velocity, $(m_i, \boldsymbol{r}_i, \boldsymbol{v}_i)$, the equations of motion for these bodies under mutual attraction from Newton's Law of Gravitation can be expressed as:

$$\begin{aligned} m_i \ddot{\boldsymbol{r}}_i &= -\frac{\partial \mathcal{U}}{\partial \boldsymbol{r}_i} \\ \mathcal{U} &= -\mathcal{G} \sum_{1 \leq j < k \leq N} \frac{m_j m_k}{\|\boldsymbol{r}_j - \boldsymbol{r}_k\|} \end{aligned}$$

where $\|\boldsymbol{r}\| = \sqrt{\boldsymbol{r} \cdot \boldsymbol{r}}$ is the Euclidian norm of a vector.

The 10 Integrals of Motion

The NBP has a number of fundamental integrals defined for it, summarized in the following:

Conservation of Linear Momentum

$$\sum_{i=1}^{N} m_i \boldsymbol{v}_i = \boldsymbol{P}$$

where $\boldsymbol{P}$ is a constant vector. This expression can be integrated once (recalling that $\boldsymbol{v}_i = \dot{\boldsymbol{r}}_i$):

$$\sum_{i=1}^{N} m_i \boldsymbol{r}_i = M \boldsymbol{R}_0 + \boldsymbol{P}(t - t_0)$$

where $M = \sum_{i=1}^{N} m_i$ is the total mass and $\boldsymbol{R}_0$ is the center of mass of the system at time t_0.

Conservation of Angular Momentum

$$\boldsymbol{H} = \sum_{i=1}^{N} m_i \boldsymbol{r}_i \times \boldsymbol{v}_i$$

where $\boldsymbol{H}$ is the total angular momentum. This expression cannot be integrated for a second time unless $N = 2$.

Conservation of Energy

$$
\begin{aligned}
E &= T + \mathcal{U} \\
T &= \frac{1}{2}\sum_{i=1}^{N} m_i \boldsymbol{v}_i \cdot \boldsymbol{v}_i
\end{aligned}
$$

and $\mathcal{U}$ is the potential energy, as defined above. T is the kinetic energy of the system, and E is the total mechanical energy of the system.

Taken together these comprise 10 integrals of motion available for any system. We note that even in the 2BP we require 12 integrals of motion to fully solve the system. The additional integral available from the angular momentum integral provides the 11th integral and the 12th integral can always be found by quadratures, and in the 2BP results in the Kepler Equation.

A convenient restatement of the angular momentum and energy integrals is possible through the application of Lagrange's Identity:

$$
\begin{aligned}
\boldsymbol{H} &= \frac{1}{M}\sum_{1\le i<j\le N} m_i m_j \boldsymbol{r}_{ij} \times \boldsymbol{v}_{ij} \\
T &= \frac{1}{2M}\sum_{1\le i<j\le N} m_i m_j \boldsymbol{v}_{ij} \cdot \boldsymbol{v}_{ij}
\end{aligned}
$$

where $\boldsymbol{r}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i$.

Elimination of Linear Momentum

The NBP can have its total degrees of freedom easily reduced by 3 by application of the conservation of linear momentum. We do this practically by solving for one body and its velocity from the 6 integrals associated with the linear momentum and writing the equations of motion independent of that body. Assume we choose to eliminate body $i = N$, meaning that we write all motion relative to this body. The resulting equations of motion are then stated as

$$
\begin{aligned}
\ddot{\boldsymbol{r}}_{Ni} &= \frac{-\mathcal{G}\left(m_i + m_N\right)}{\|\boldsymbol{r}_{Ni}\|^3}\boldsymbol{r}_{Ni} - \mathcal{G}\sum_{j=1, j\neq i}^{N-1} m_j \left[\frac{\boldsymbol{r}_{ji}}{\|\boldsymbol{r}_{ji}\|^3} - \frac{\boldsymbol{r}_{Nj}}{\|\boldsymbol{r}_{Nj}\|^3}\right] \\
j &= 1, 2, \ldots, N-1
\end{aligned}
$$

where $\boldsymbol{r}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i$. For $N = 2$ this simplifies to the usual statement of the 2BP. Once the above system is solved, the motion of body P_N can then be found from the conservation of linear momentum.

Lagrange-Jacobi Identity

Define the polar moment of inertia as

$$
\begin{aligned}
I &= \sum_{i=1}^{N} m_i \boldsymbol{r}_i \cdot \boldsymbol{r}_i \\
&= \frac{1}{M}\sum_{1\le i<j\le N} m_i m_j \boldsymbol{r}_{ij} \cdot \boldsymbol{r}_{ij}
\end{aligned}
$$

Taking the 2nd time derivative of I we find the Lagrange-Jacobi Identity:

$$\begin{aligned}\ddot{I} &= 4T + 2\mathcal{U} \\ &= 2T + 2E \\ &= -2\mathcal{U} + 4E\end{aligned}$$

where all of the different statements are equivalent, given conservation of energy. This result rests on the fact that U is homogenous of degree -1.

Sundman's Inequality

Application of the Cauchy Inequality to the angular momentum provides an important constraint between angular momentum, kinetic energy, and the system moment of inertia. As seen in the homework, we find the result:

$$H^2 \leq 2TI$$

This can be used to find a lower-bound on the energy of an N-body system for a given value of angular momentum

$$E \geq \frac{H^2}{2I} + \mathcal{U}$$

Theorems on Motion in the NBP

Consider the following definitions:

Hill Stability A system is Hill Stable if $\|\boldsymbol{r}_{ij}\| \leq C < \infty$ for all i, j and for all time.

Lagrange Stability A system is Lagrange Stable if $0 < c \geq \|\boldsymbol{r}_{ij}\| \leq C < \infty$ for all i, j and for all time.

Total Collapse A total collapse of a system occurs if $\|\boldsymbol{r}_{ij}\| \to 0$ for all i, j at the same time.

We can prove the following Theorems:

Necessary Condition for Hill Stability A necessary condition for Hill stability is that $E < 0$.

Sufficient Condition for Hill Instability A sufficient condition for Hill instability is $E > 0$, with at least one body escaping to infinity.

Sundman's Theorem on Total Collapse If total collapse occurs, then $H = 0$ and collapse only takes a finite time, i.e., occurs at some time $-\infty < t_C < \infty$.

Minimum Energy Configurations The minimum energy configuration for a system with a fixed angular momentum H and $N \geq 3$ is unbounded from below. This implies that a system with ≥ 3 Hill-Stable components that dissipates energy will eventually result in a collision between two of the bodies. For $N = 2$ there always exists a minimum energy orbit for a given value of H, corresponding to a circular orbit.

Non-existence of Minimum Energy Configurations For $N \geq 3$ there are no central configurations that are also minimum energy configurations for a given value of angular momentum H. Thus, for $N \geq 3$ there are no local energy minima, meaning that all such systems that dissipate energy will end in a collision.

Central Configurations

A necessary and sufficient condition for a collection of N particles to form a central configuration is that

$$\frac{\partial(I\mathcal{U}^2)}{\partial \boldsymbol{r}_i} = 0$$

for $i = 1, 2, \ldots, N$.

Similary, a necessary and sufficient condition for a central configuration with a specified level of angular momentum H is that

$$\frac{\partial}{\partial \boldsymbol{r}_i}\left[\frac{H^2}{2I} + \mathcal{U}\right] = 0$$

for $i = 1, 2, \ldots, N$.

Homothetic Solutions A homothetic solution is one which always remains similar to itself over with respect to a uniform scaling only. A homothetic solutions always has $H = 0$ and will either suffer a total collapse or a total escape. A central configuration can always be made into a homothetic solution.

Homographic Solutions A homographic solution is one which always remains similar to itself over time by either a rotation or a uniform scaling. Any planar central configuration can be made into a homographic solution. If a central configuration is non-planar, then the only homographic solutions it can have are homothetic solutions. The uniform scaling and rotation rate parameters of a homographic solution follow a set of differential equations that are equivalent to the reduced 2BP:

$$\begin{aligned}
\ddot{\alpha} - \alpha\Omega^2 &= \frac{\sigma^*}{\alpha^2} \\
\alpha\dot{\Omega} + 2\dot{\alpha}\Omega &= 0 \\
\sigma^* &= \frac{\mathcal{U}^*}{I^*}
\end{aligned}$$

where α is a scaling parameter (like a normalized radius) and Ω is a rotation rate about a fixed axis perpendicular to the central configuration plane. The parameters $\mathcal{U}^*$ and I^* are the force potential and moment of inertia evaluated at the central configuration. Given a central configuration $\boldsymbol{r}_i^*$, the homographic solution is then:

$$\begin{aligned} \boldsymbol{r}_i &= \alpha \boldsymbol{r}_i^* \\ \boldsymbol{v}_i &= \dot{\alpha} \boldsymbol{r}_i^* + \alpha \Omega \hat{\boldsymbol{\Omega}} \times \boldsymbol{r}_i^* \end{aligned}$$

where $\hat{\boldsymbol{\Omega}}$ is the unit vector along the rotation axis.

The total energy and angular momentum of a homographic solution can be expressed as:

$$\begin{aligned} E &= \left[\frac{1}{2} \left(\dot{\alpha}^2 + \alpha^2 \Omega^2 \right) + \frac{\sigma^*}{\alpha} \right] I^* \\ \boldsymbol{H} &= \alpha^2 \Omega I^* \hat{\boldsymbol{\Omega}}. \end{aligned}$$

Full 2-Body Problem

For the F2BP we find the following definitions of kinetic energy, potential energy, angular momentum, and moment of inertia:

$$\begin{aligned}
T &= \frac{1}{2}\sum_{I=1}^{2} \boldsymbol{\Omega}_I \cdot \boldsymbol{\mathcal{I}}_I \cdot \boldsymbol{\Omega}_I + \frac{1}{2}\frac{m_1 m_2}{m_1+m_2}\boldsymbol{v}\cdot\boldsymbol{v} \\
\mathcal{U} &= \mathcal{U}_{11} + \mathcal{U}_{22} + \mathcal{U}_{12} \\
\mathcal{U}_{12} &= -\mathcal{G}\int_{\mathcal{B}_1}\int_{\mathcal{B}_2} \frac{dm_1 dm_2}{\|\boldsymbol{r} + \boldsymbol{A}_1\cdot\boldsymbol{\rho}_1 - \boldsymbol{A}_2\cdot\boldsymbol{\rho}_2\|} \\
\boldsymbol{H} &= \sum_{I=1}^{2} \boldsymbol{A}_I \cdot \boldsymbol{\mathcal{I}}_I \cdot \boldsymbol{\Omega}_I + \frac{m_1 m_2}{m_1+m_2}\boldsymbol{r}\times\boldsymbol{v} \\
I_H &= \hat{\mathbf{H}}\cdot[\boldsymbol{\mathcal{I}}_1 + \boldsymbol{\mathcal{I}}_2]\cdot\hat{\mathbf{H}} + \frac{m_1 m_2}{m_1+m_2}\left(r^2 - (\boldsymbol{r}\cdot\hat{\mathbf{H}})^2\right)
\end{aligned}$$

we note that the Sundman Inequality holds unchanged.

Minimum Energy Configurations

For a given value of angular momentum, H, a lower bound on the energy is:

$$\frac{H^2}{2I_H} + \mathcal{U} \leq E$$

Stationary values of this lower bound are central configurations (in the N-body problem) or relative equilibria (for the F2BP). Global or local minima of this function can only be found for the 2BP and the F2BP, respectively. The structure of the energy curves between these two problems is significantly different, with the 2BP only having one global minima and the F2BP having both a local minimum and a local maximum.